Assignment 4

Congratulations! You have learnt about how to create a HTTP server with Go. Unfortunately your application is not secure enough, there are several things that you need to do:

1. Create user data with these attributes: ID, Name, Username, Password
2. Seed the user data with minimum 1 user, the password must be hashed in BCRYPT 10 round.
3. Create Login Endpoint
4. Create Auth Middleware which will validate if the JWT token is valid and is not expired. The JWT token will be passed inside Authorization header as Bearer token.
5. Create Get Current Authenticated User endpoint

a. Login Endpoint

Description

Create a login endpoint which will take username and password. Make sure to verify the request password with the hashed password in the user data store. If the user is found and verified, return the JWT Token with id, name, username as the token claims.

Method

POST

Endpoint

/api/auth/login

Request Body

Attribute	Data Type
username	string
password	string

Request Sample

```
{
    "username": "user1",
    "password": "StrongPassword"
}
```

Response Body

Success Response 200

Attribute	Data Type
accessToken	string

Success Response 200 Sample

```
{
    "accessToken":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MSwibmFtZSI6IlVzZXIgMSIsInVzZXJuYW1lIjoidXNlcjEifQ.nIrfL6knF9GV_2zfO0S5
jVL7czElDu6ywFMnt7KpR_U"
}
```

Unauthorized Response 401

Attribute	Data Type
error	string

Unauthorized Response 401 Sample

```
{
    "error": "Unauthorized"
}
```

b. Get Current Authenticated User Endpoint

Description

Create an endpoint which will retrieve the current user data from the JWT token.

Method

GET

Endpoint

/api/auth/me

Request Header

Attribute	Data Type	Sample
Authorization	string	Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZCI6MSwibmFtZSI6IlVzZXIgMSIsInVzZXJuYW1lIjoidXNlcjEifQ.nIrfL6knF9GV_2zfO0S5jVL7czEID

Response Body

Success Response 200

Attribute	Data Type
id	integer
username	string
name	string

Success Response 200 Sample

```
{
    "id": 1,
    "username": "user1",
    "name": "User 1"
}
```

Unauthorized Response 401

Attribute	Data Type
error	string

Unauthorized Response 401 Sample

```
{
    "error": "Unauthorized"
}
```